

Prediction of Continuous B-cell Epitopes Using Long Short Term Memory Networks

Bin Cheng

Institute of Applied
Mathematics, Hebei
Academy of Sciences
Hebei Authentication
Technology Engineering
Research Center,
Shijiazhuang, HB, China
46 Youyi South Street,
Shijiazhuang, Hebei,
China
+86-311-67306732
chengbin@heb_mat
h.org

Ling-yun Liu

College of Information
Science and Technology,
Shijiazhuang Tiedao
University
17 Northeast, Second
Inner Ring, Shijiazhuang,
Hebei, China
+86-311-673067
1272601021@qq.co
m

Zhao-hui Qi

College of Information
Science and Technology,
Shijiazhuang Tiedao
University
17 Northeast, Second
Inner Ring, Shijiazhuang,
Hebei, China
+86-311-87935049
zhqi_wy2013@163.c
om

Hong-guang Yang

Institute of Applied
Mathematics, Hebei
Academy of Sciences
46 Youyi South Street,
Shijiazhuang, Hebei,
China
+86-311-67306732
yanghongguang@he
b-math.org

ABSTRACT

B-cell epitopes play a vital role in the epitope-based vaccine design. The accumulation of epitope sample data makes it possible to predict epitopes using machine learning methods. Compared with the experimental tests, the computational methods are faster and more economic. Several machine learning computational methods have been applied to improve the accuracy of epitope predictions. These methods have been improved several times in the epitope prediction has made some achievements, but there are also deficiencies. The commonly used propensity scale methods for the prediction are physicochemical properties of amino acid sequences. It is difficult to get a good classification result in the network training using only the physicochemical properties of the sample sequence. In this study, we have developed a novel method for predicting continuous B-cell epitope. We adopted the Long Short Term Memory network and relevance of amino acids pair feature scale. Long Short Term Memory network can make up for the lack of recurrent artificial neural network algorithm, which is very suitable for epitope prediction. We have been adopted the performance of Long Short Term Memory network and the relevance of amino acids pair feature scale in three aspects, and achieved a certain result.

CCS Concepts

• Applied computing → Bioinformatics

Keywords

B-cell epitopes; Prediction; LSTM Network; RAAP; Relevance of amino acids pair.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from Permissions@acm.org.

ICBCB 2018, March 12–14, 2018, Chengdu, China

© 2018 Association for Computing Machinery.

ACM ISBN 978-1-4503-6348-8/18/03...\$15.00

<https://doi.org/10.1145/3194480.3194493>

1. INTRODUCTION

A B-cell epitope is the binding portion of an antigen that interacts with a B-cell receptor or an antibody molecule [1]. When an antigen is a protein, an epitope is a short peptide from the protein sequence. B-cell epitopes get the extensive application as key role in the designing of peptide-based vaccines and also in the diagnosis of diseases. Continuous epitopes are the kind of B-cell epitopes and formed by continuous amino acid sequences. Another type of B-cell epitopes is discontinuous epitopes, which are composed of distantly separated residues in the protein sequence [2-4].

Thirty years ago, epitope predictions were based on experimental tests that took a long time and costly [5-8]. Artificial intelligence technology provides a rapid and economical method for epitope prediction, which provides a good reference for understanding the epitope of protein [9-10]. Several machine learning computational methods have been applied to improve the accuracy of epitope predictions. Among them, support vector machines (SVM) and recurrent artificial neural network (RNN) are more used. For example, Manoj [11] proposed a models to discriminate epitopes and non-epitopes using quantitative matrix, SVM and Artificial Neural Network (ANN). BEEPro [12] adapts a linear averaging scheme on 16 properties using a support vector machine model to predict both linear and conformational B-cell epitopes. ABCPred [13] uses RNN for predicting linear B-cell epitopes and was evaluated on a dataset of 700 B-cell epitopes and 700 non-epitope peptides using 5-fold cross validation tests. These methods have been improved several times in the epitope prediction has made some achievements, but there are also deficiencies. Such as, RNN network in the case of a large number of layers disappear gradient and explosion problems.

The commonly used propensity scale methods for the prediction are physicochemical properties of amino acid sequences, such as hydrophilicity [14], flexibility [15], accessibility [16], turns [17], and the like. Of course there are other propensity scales for epitope prediction. Chen et al. [18] developed an amino acid pair (AAP) antigenicity scale that assigns to each possible pair of amino acids and trained a SVM classifier. LEPS [19] adopted the library for SVM tool and trained it to recognize features of amino acid segments (AASs) with lengths from 2 to 4 residues. BEEPro adapts a linear averaging scheme on 16 properties that include

position specific scoring matrix (PSSM), an amino acid ratio scale, and a set of 14 physicochemical scales obtained via a feature selection process. These methods all improve the accuracy of prediction.

Several types of feature selection methods have been developed to address the epitope predictions problem. In general, feature selection methods consider the relevance and diversity among the feature [20]. It is difficult to get a good classification result in the network training using only the physicochemical properties of the sample sequence. Therefore, sequence features are constructed as a combination of physicochemical properties from the sample sequence and their associated combinations.

Epitope sequences are fragments of the primary sequence of proteins and belong to a type of text that has a dependency relationship between the data. Text classification generally combines forward and backward to form a bidirectional feature, and each cell has a state feature and an output feature [21]. RNN is suitable for classification of epitopes, but sometimes there is a problem that the error return is lost [22]. Long Short Term Memory (LSTM) network can make up for the lack of RNN algorithm, which is very suitable for epitope prediction.

In this study, we have developed a novel method for predicting continuous B-cell epitope. We adopted the LSTM network and the relevance of amino acids (RAAP) feature scale. The LSTM network with a variety of optimization methods to achieve better distinction between epitopes and non-epitopes. The RAAP feature scale focuses on the information relevance of adjacent amino acids as well make sure the learning characteristics without excessive differences.

2. Materials AND Methods

2.1 The Data Set

In this study, two datasets were used. One is self-built dataset, the other is Chen dataset [18]. In our dataset, B-cell epitopes have been obtained from IDEB database that contains 3780 continuous epitopes and 3780 non-epitopes were generated by selecting peptide segments from the UniProt database. To meet the training requirements, we have fixed length pattern whereas B-cell epitopes have varying length [13]. For the epitope length is less than 20 amino acids, we have increased its length by introducing equal number of residues at both terminals derived from its original antigenic sequence. The Chen dataset contained 872 epitopes and 872 non-epitopes. All epitopes and non-epitopes within this dataset were restricted to a length of 20 residues.

2.2 Prediction algorithm

In this study, SVM, ELMAN and LSTM have been used. In SVM each sample is represented by a vector. The SVM classifier basically learns how to classify every query sample into Classe 1 or Classe -1 from a set of labeled training examples. The ELMAN network is a structural type of RNN network, and it contains feedback connections from implicit nodes to contextual nodes. The ELMAN network can analyze the relationship between the time series using the idea of weight-sharing. However, SVM and ELMAN did not yield satisfactory result and prompted us to try for LSTM network.

The LSTM network is an improved model developed on the basis of RNN. As with most RNN, the LSTM network can complete any calculations that traditional computers can do in theory, as long as they have enough basic units and appropriate weighting matrices. Different from the problem of exponential error

attenuation of RNN, the LSTM error propagates in the first order without any attenuation or explosion problems. The LSTM network sets up a memory block network that spans the storage of information beyond any delay. The LSTM network can return the error signal to a long time ago and achieve satisfactory results in epitope prediction.

In this study, a fivefold cross-validation technique has been used, in which the two datasets is randomly divided into five subsets respectively, each containing an equal number of peptides. These subsets are used as training, verification and testing. The final prediction results have been the average of five testing sets.

2.3 RAAP Feature Scale

We introduce the relevance of amino acids pair (RAAP) which is a novel feature scale. RAAP are generated by decomposing the epitope sequences. Any adjacent three amino acids in the epitope sequence can be decomposed into an amino acids pair (AAP). For example, the epitope sequence MCTGSFKLEK can be decomposed into 8 AAPs: MCT, CTG, TGS, GSF, SFK, FKL, KLE, and LEK. The RAAP feature scale is defined by

$$RAAP = \left(\sum_{i=1}^3 PV_{AAP_i} \right) / 3 \quad (1)$$

Where PV_{AAP_i} is the i th amino acid in AAP and PV is the value of physicochemical properties of amino acid. The RAAP feature scale will reflect the relevance feature information of amino acids pair and avoid the dominance of an individual feature in the classifier learning.

2.4 Evaluation Parameters

As is known, five indicators were used to measure the performance of prediction model. The first is sensitivity (SEN), which is defined as the percentage of epitopes that were correctly predicted as epitopes. The second is specificity (SPE), which is the percentage of non-epitopes that were correctly predicted as non-epitopes. The third is positive predictive value (PPV), which is used to stand for the probability that a true epitope was predicted as an epitope. The forth is accuracy (ACC), which refer to the proportion of properly predicted epitopes. The last one is Matthews' correlation coefficient (MCC), which is the value that incorporated both SEN and SPE into a single value between -1 and +1. These parameters can be calculated with the following equations:

$$SEN = \frac{TP}{TP + FN} * 100\% \quad (2)$$

$$SPE = \frac{TN}{TN + FP} * 100\% \quad (3)$$

$$PPV = \frac{TP}{TP + FP} * 100\% \quad (4)$$

$$\text{ELMAN ACC} = \frac{TP + TN}{TP + FP + TN + FN} * 100\% \quad (5)$$

$$\text{MCC} = \frac{TP * TN - FP * FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (6)$$

where TP and TN are the number of correctly predicted epitope, and non-epitope residues, respectively, FP is the number of non-epitope sequences that were predicted to be in epitopes, and FN is the number of epitope sequences that were predicted not to be in epitopes[23].

3. RESULTS AND DISCUSSION

3.1 Performance of Models Based on Physico-chemical Properties

To evaluate the performance of LSTM, we compute its predictive performance in three aspects. First of all, we have adopted hydrophilicity of amino acids as an input feature using our dataset, and introduced the feature scale using RAAP. All the methods have been trained and tested using fivefold cross-validation. The prediction performance measures have been averaged over five sets. We implemented three methods (Table 1) for training and testing, and achieved accuracy in the range of 65.5–88.7% with MCC of 42.01%–77.44%. In addition, We implemented three methods using Chen dataset in the same conditions (Table 2). We observed ACC of 54.36, 80.50 and 88.86 for SVM, ELMAN and LSTM respectively. In summary, LSTM prediction model performs better than SVM and ELMAN in both datasets.

Table 1. The performance of three methods developed on hydrophobicity of amino acids using our dataset

Method	SEN (%)	SPE (%)	PPV (%)	ACC (%)	MCC
SVM	31.79	99.23	97.64	65.51	0.4202
ELAMN	55.51	99.94	99.76	87.11	0.6843
LSTM	93.08	85.90	80.85	88.70	0.7744

Table 2. The performance of three methods developed on hydrophobicity of amino acids using Chen dataset

Method	SEN (%)	SPE (%)	PPV (%)	ACC (%)	MCC
SVM	63.56	76.48	70.06	72.54	0.4040
ELAMN	61.12	99.88	99.81	80.50	0.6618
LSTM	99.19	74.52	84.39	88.86	0.7818

Secondly, we have adopted Hydrophilic and hydrophobic and the feature scale RAAP as the input feature using two datasets. We have applied three types of methods for trained and tested using fivefold cross-validation. As shown in Table 3, for LSTM prediction model in our dataset, the average sensitivity in the dataset reached 99.19%, the ACC reached 88.86%, the MCC reached 78.18%, but the SPE and PPV is lower than ELMAN method. These results indicate that the trained network prediction performance is not balanced using LSTM prediction model. Similarly, We have applied three types of methods in Chen

dataset. LSTM prediction model indicators are superior to the other two methods (Table 4).

Table 3. The performance of three methods developed on combination of two characters of amino acids using our dataset

Method	SEN (%)	SPE (%)	PPV (%)	ACC (%)	MCC
SVM	32.05	98.97	96.89	65.51	0.4175
ELAMN	56.79	99.94	99.77	87.48	0.6938
LSTM	98.46	85.57	81.36	90.6	0.8210

Table 4. The performance of three methods developed on combination of two characters of amino acids using Chen dataset

Method	SEN (%)	SPE (%)	PPV (%)	ACC (%)	MCC
SVM	50.42	56.50	57.76	53.21	0.0691
ELAMN	55.05	61.24	58.68	58.14	0.1632
LSTM	98.27	67.36	80.69	85.33	0.7121

Thirdly, we trained and tested LSTM prediction model using two datasets for different window lengths (input units) like 10, 12, 14, 16, 18, and 20. As shown in Table 5, the accuracy of the LSTM prediction method gets better as the window gets bigger and similar to references 13.

Table 5. The performance of LSTM using hydrophobicity of amino acids with different window sizes

Method	SEN (%)	SPE (%)	PPV (%)	ACC (%)	MCC
10	41.84	63.27	47.69	53.75	0.0522
12	40.68	62.66	51.80	51.75	0.0343
14	42.11	64.96	60.51	52.15	0.0719
16	41.24	65.69	71.54	49.15	0.0665
18	40.82	63.69	62.44	50.05	0.4541
20	80.84	95.09	93.07	88.70	0.7744

3.2 Testing of RAAP Feature Scale

To evaluate the efficiency of RAAP feature scale, we developed SVM prediction model using Chen dataset. We then compare the performance of SVM classifier using RAAP feature scale with that using AAP antigenicity scale. The AAP antigenicity scale is occurrence frequencies of some AAPs. As shown in Table 6, the performance of SVM using RAAP is better than using AAP

Table 6. The performance of SVM methods using AAP antigenicity scale and RAAP feature scale on Chen dataset

Method	SEN (%)	SPE (%)	PPV (%)	ACC (%)	MCC
AAP antigenicity scale ^[18]	60.87	75.36	68.07	71.09	0.3663
RAAP feature scale	63.56	76.48	70.06	72.54	0.4040

3.3 Discussion

Many algorithms have been developed to predict continuous B-cell epitopes and obtained some achievements. However, the accuracy of epitope predictions has not been satisfactory. In general, epitope prediction is looking for more effective learning algorithms on the one hand, and more effective learning features on the other hand.

In this study, for the first time a systematic attempt has been made to predict B-cell epitopes based on LSTM network. On the one hand, we need a solutions to solve the problem of gradient instability in deep learning. LSTM method is a good solution, and suitable for the classification of text sequences. SVM and RNN are more used to improve the accuracy of epitope predictions and achieve a certain result. However, LSTM prediction model can get better classification performance. The causes are those network model have single structure and have no advantage in text classification. On other hand, a major problem in machine learning method is feature selection. Relevance of amino acids pair is a novel feature scale and can reflect the relevance feature information of amino acids pair and avoid the dominance of an individual feature in the classifier learning.

Though we have obtained high prediction accuracy of B-cell epitopes in our study, it has some disadvantages. In the case of the training results, network optimization still had the existence space.

4. CONCLUSIONS

Prediction of the B-cell epitopes is an important but also very difficult and complicated problem. We came up with a novel feature scale, also referred to as RAAP, to deal with the problem. The RAAP scale approach can give much better performance than AAP scale approaches for B-cell epitope prediction based on SVM method. Simultaneously, we have introduced LSTM network for prediction of continuous B-cell epitopes. We have compared the performance of LSTM prediction model on several conditions. It is also observed that LSTM prediction model performed better than the other models developed on two type datasets.

5. ACKNOWLEDGMENTS

This work was supported by Cooperation Science & Technology program of Hebei Academy of Sciences and Chinese Academy of Sciences [171406]. All authors contribute the same to this article. Corresponding author is Yang hongguang.

6. REFERENCES

- [1] Regenmortel M.H.V.V. 2009. What is a B-cell epitope? *Epitope Mapping Protocols*. 524: 3-20. DOI= https://doi.org/10.1007/978-1-59745-450-6_1
- [2] Ren J., Liu Q., Ellis J., Li J.Y. 2015. Positive-unlabeled learning for the prediction of conformational B-cell epitopes. *BMC Bioinformatics*. 16(S18): S12. DOI= <https://doi.org/10.1186/1471-2105-16-S18-S12>
- [3] Hu Y.J., Lin S.C., Lin Y.L., Lin K.H., You S.N. 2014. A meta-learning approach for B-cell conformational epitope. *BMC Bioinformatics* 15(1): 1-15. DOI= <https://doi.org/10.1186/s12859-014-0378-y>
- [4] Zheng W., Zhang C., Hanlon M., Ruan J.S., Gao J.Z. 2014. An ensemble method for prediction of conformational B-cell epitopes from antigen sequences. *Computational Biology and Chemistry*. 49: 51-58. DOI= <https://doi.org/10.1016/j.cmpbiolchem.2014.02.002>
- [5] Muller S., Benkirane N., Guichard G., Van Regenmortel M.H.V., Brown F. 1998. The potential of retro-inverso peptides as synthetic vaccines. *Expert Opinion on Investigational Drugs* 7(9): 1429-1438. DOI= <https://doi.org/10.1517/13543784.7.9.1429>
- [6] Muller G.M., Shapira M., Arnon R.. 1982. Anti-influenza response achieved by immunization with a synthetic conjugate. *Proc Natl Acad Sci USA* 79(2): 569-573. DOI= <https://doi.org/10.1073/pnas.79.2.569>
- [7] Hopp T.P., Woods K.R. 1981. Prediction of protein antigenic determinants from amino acid sequences. *Proc. Natl. Acad. Sci. USA* 78(6):3824-3828. DOI= <https://doi.org/10.1073/pnas.78.6.3824>
- [8] Levitt M. 1976. A simplified representation of protein conformations for rapid simulation of protein folding. *Journal of Molecular Biology* 104(1): 59-107. DOI= [https://doi.org/10.1016/0022-2836\(76\)90004-8](https://doi.org/10.1016/0022-2836(76)90004-8)
- [9] Yao B., Zheng D.D., Liang S.D., Zhang C. 2013. Conformational B-Cell Epitope Prediction on Antigen Protein Structures: A Review of Current Algorithms and Comparison with Common Binding Site Prediction Methods. *PLOS ONE* 8(4):e62249. DOI= <https://doi.org/10.1371/journal.pone.0062249>
- [10] Singh H., Ansari H.R., Raghava G.P.S. 2013. Improved Method for Linear B-Cell Epitope Prediction Using Antigen's Primary Sequence. *PLOS ONE* 8(5): e62216. DOI= <https://doi.org/10.1371/journal.pone.0062216>
- [11] Bhasin M., Raghava G.P. 2004. Prediction of CTL epitopes using QM, SVM and ANN techniques. *Vaccine* 22(22-23): 3195-3204. DOI= <https://doi.org/10.1016/j.vaccine.2004.02.005>
- [12] Lin S.Y., Cheng C.W., Su E.C.Y. 2013. Prediction of B-cell epitopes using evolutionary information and propensity scales. *BMC Bioinformatics* 14(S2): S10. DOI= <https://doi.org/10.1186/1471-2105-14-S2-S10>
- [13] Saha S., Raghava G.P.S. 2006. Prediction of continuous B-cell epitopes in an antigen using recurrent neural network. *PROTEINS: Structure, Function, and Bioinformatics* 65(1): 40-48. DOI= <https://doi.org/10.1002/prot.21078>
- [14] Parker J.M., Guo D., Hodges R.S. 1986. New hydrophilicity scale derived from high-performance liquid chromatography peptide retention data: correlation of predicted surface residues with antigenicity and X-ray-derived accessible sites. *Biochemistry* 25(19): 5425-5432. DOI= <https://doi.org/10.1021/bi00367a013>
- [15] Lian Y., Ge M., Pan X.M. 2014. EPMLR: sequence-based linear B-cell epitope prediction method using multiple linear regression. *BMC Bioinformatics* 15:414. DOI= <https://doi.org/10.1186/s12859-014-0414-y>
- [16] Rahman K.S., Chowdhury E.U., Sachse K., Kaltenboeck B. 2016. Inadequate Reference Datasets Biased towards Short Non-epitopes Confound B-cell Epitope Prediction. *Journal of Biological Chemistry* 291(28): 14585. DOI= <https://doi.org/10.1074/jbc.M116.729020>
- [17] Kozlova E.E.G., Viart B.T., Avila R.A.M., Felicori L.F., Chavez-Olortegui C. 2015. Classification epitopes in groups based on their protein family. *BMC Bioinformatics* 16(S19):S7. DOI= <https://doi.org/10.1186/1471-2105-16-S19-S7>

- [18] Chen J., Liu H., Yang J., Chou K.C. 2007. Prediction of linear B-cell epitopes using amino acid pair antigenicity scale. *Amino Acids* 33(3): 423-428. DOI= <https://doi.org/10.1007/s00726-006-0485-9>
- [19] Wang H.W., Lin Y.C., Pai T.W., Chang H.T. 2011. Prediction of B-cell Linear Epitopes with a Combination of Support Vector Machine Classification and Amino Acid Propensity Identification. *Journal of Biomedicine and Biotechnology* 2011: 432830. DOI= <https://doi.org/10.1155/2011/432830>
- [20] Ghalwash M.F., Cao X.H., Stojkovic I., Obradovic Z. 2016. Structured feature selection using coordinate descent optimization. *BMC Bioinformatics* 17: 158. DOI= <https://doi.org/10.1186/s12859-016-0954-4>
- [21] Song M., Zhao Y., Wang S. 2017. Exploiting different word clusterings for class-based RNN language modeling in speech recognition. *IEEE International Conference on Acoustics* 2017: 5735-5739. DOI= <https://doi.org/10.1109/ICASSP.2017.7953255>
- [22] Donahue J., Hendricks L.A., Guadarrama S., Rohrbach M., Venugopalan S., Darrell T., Saenko K. 2015. Long-term recurrent convolutional networks for visual recognition and description. *Computer Vision & Pattern Recognition* 239(4): 677-691. DOI= <https://doi.org/10.1109/CVPR.2015.7298878>
- [23] Sun P.P., Qi J.L., Zhao Y.Z., Huang Y.X., Yang G.F., Ma Z.Q., Li Y.X. 2016. A novel conformational B-cell epitope prediction method based on mimotope and patch analysis. *Journal of Theoretical Biology*. 394: 102-108. DOI= <https://doi.org/10.1016/j.jtbi.2016.01.021>